

Overseeing Many LLMs: A User-Controlled Review Queue for Concurrent Agent Workflows

Anonymous Submission

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Abstract: Concurrent language model workers can produce more requests than one user can immediately inspect. We present a review workflow that preserves active evidence during arrivals, keeps unfinished work visible and associates release authority with an exact output version. The implemented desk supports worker conversations, a central queue and optional source-based sessions through one asynchronous event protocol. A frozen study uses 24 fresh TAT-QA contexts, 145 original questions and two generated-answer replicas. The model produces 61 exact matches among 290 answers, with 38 empty outputs retained. Across 144 matched wall-clock software scenarios, all 13,236 events replay consistently and active-review checks detect no unintended changes. All three interfaces finish the ideal scripted workload, so completion ties mechanically. Source sessions do not demonstrate an efficiency advantage. We distinguish these software results from human effectiveness and provide disjoint task packets, training and a prospective counterbalanced study. The contribution is a specified and executable oversight workflow with evidence about its concurrency and release behavior, not a claim of reduced human workload.

1 INTRODUCTION

A person supervising several language model workers needs to know which outputs require a decision, what evidence supports them, and what remains unfinished. An arriving answer should not replace the source they are reading. A postponed question should retain its draft and history. If an agent revises an answer after approval, the new artifact should not silently inherit permission to be released. These are concrete requirements for maintaining control of concurrent work.¹

We investigate how a person can organize, inspect and resolve competing actionable requests while retaining visibility of unfinished work and control of release. Our application is financial question answering. Several independently prompted task workers use one shared model to answer original questions about real financial-report tables. The user can inspect the source, approve one output, supply a correction, reject it or defer it. A shared source supports navigation, but never implies that two questions share a correct answer.

The design favors a strong simple queue over elaborate allocation or automatic correction propagation. Earlier internal studies found that more complicated ordering and uncertain correction reuse did not consistently improve outcomes. Those findings motivate

separate decisions, direct evidence and explicit user control. They do not validate this interface.

Our contributions are a precise review-and-release protocol, a working desk with live and saved-output adapters, and a frozen real-data software evaluation. The protocol makes accounting, focus stability and version authority testable under concurrent arrivals. A prospective human study compares the central queue with optional source sessions while keeping fundamental approval semantics common. We present this as a position supported by an implemented system and measured software behavior. Human review quality and experienced workload remain open empirical questions.

2 RELATED WORK

Agent dashboards and intervention interfaces already have close precedents. AgentGUI supports concurrent desks, activity views, intervention and saved sessions. Its eight-participant evaluation concerns trace comprehension, while a separate experiment tests automated steering (Zhao et al., 2026). Our evaluation concerns actionable output requests and exact-version release, not a reproduction of its user study.²

¹AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

²AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

AGDebugger provides queues, pause/step controls, message edits and checkpointed conversation forks. Its studies examine error identification and steering of multi-agent teams (Epperson et al., 2025). These mechanisms establish that queues, history and intervention are not new contributions by themselves. Our narrower focus is the lifecycle of individual outputs awaiting a user’s release decision.

Human multi-robot supervision also has a substantial history of attention allocation. Chien et al. compare open-queue and shortest-job-first presentation (Chien et al., 2018). Research on interruption recovery motivates retaining a postponed task’s context (Iqbal and Horvitz, 2007). We apply these established principles without interpreting queue length, token counts or scripted action time as mental demand. The proposed contribution is a specified workflow and its implementation and evaluation, not priority for an oversight interface or a new scheduling rule.

3 REVIEW PROTOCOL

3.1 Requests, Sources and Versions

An offered task records a worker identifier, original question, source identity and hash, provenance, and any explicit dependencies. An actionable request is created when its proposed answer arrives. It records the artifact version, receipt time, requested decision and review status. Generated explanations and citations are labeled as generated. The original source table and paragraphs remain available even when an answer or its citation format is incomplete.³

The task lifecycle is offered, admitted, started and received. A received request is queued, active, deferred, resolved or withdrawn. Artifacts are immutable version records. A revision supersedes the current version without deleting earlier artifacts, decisions or release history. The application does not infer dependencies from similar wording. In the TAT-QA application, source identity is used only to organize related questions.

No financial deadlines are invented. The protocol accepts a deadline or priority only with an explicit origin, such as a user’s instruction. Without such metadata, the next-request recommendation is stable FIFO. Explicit deadlines use EDF within priority. Once a review begins, the active artifact is pinned

until the user acts. The experimental observation cut-off is not a business processing deadline.

3.2 Actions and Authority

Approval authorizes one exact request/version. Release is a separate action and requires approval of the current version. A user correction creates and approves a new version of that answer only. Rejection blocks the answer and resolves the review, but does not complete the underlying question correctly. Deferral retains the source, draft, note and history, with an optional return time. Withdrawal is explicit and does not erase prior events or undo any previously recorded release.

When a revised artifact arrives during review, the old source and answer remain pinned. A visible warning offers an explicit refresh. An attempt to approve the older version is refused. After refresh, the new version requires its own decision. A group contains at most three already received questions with the same source identifier. Grouping never combines their answer fields or approval authority.

Admission control pauses new task starts. It does not interrupt an in-flight model request or prevent that request’s result from arriving. All offered tasks remain in the outstanding-work account, including those that have not started. This distinction prevents an apparently smaller queue from concealing delayed work.

3.3 Atomic Events and Limited Properties

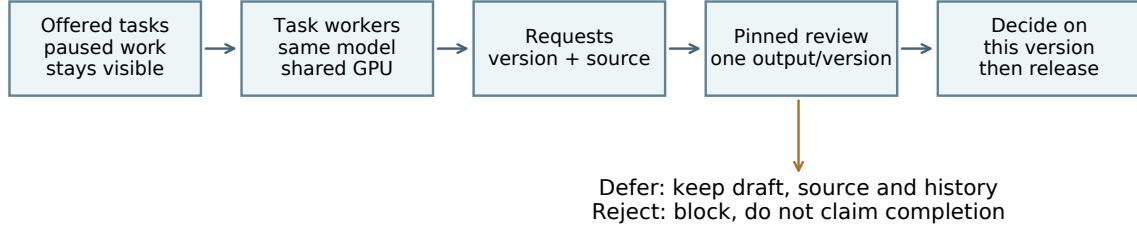
Commands are processed under one lock. Each includes an event ID, action and payload. A repeated ID with identical content returns the earlier receipt. Reusing an ID for different content is refused. Before applying a transition, the controller checks its prerequisites. A refused transition restores the prior logical state. Each command is recorded with a sequence, timestamps, result and state hash.

Let O be offered tasks, U unstarted tasks, F in-flight tasks and R received requests. The implementation checks

$$O = U + F + R. \quad (1)$$

Received requests are similarly partitioned into queued, active, deferred, resolved and withdrawn states. No accepted transition deletes a request. By induction over transitions, every offered task therefore remains accounted for. Immutable version records and copied active snapshots imply that a new arrival cannot change the active reviewed artifact. A release

³AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.



New arrivals update pending work. A revision never inherits an approval.

Figure 1: Specified software workflow. Every output has a separate version and release decision. Source sessions change navigation, not answer authority. This diagram describes the protocol rather than a measured human effect.

check requires the current version and its own approving decision, so a revision cannot inherit approval. Grouping changes only membership and cannot authorize a sibling output.

These are structural properties under the implemented protocol. They do not imply answer correctness, user comprehension or low workload. All releases must pass through this controller. The local single-user server is not a distributed consensus service or an authenticated public deployment. Journals support recovery and replay, but are not claimed as an fsync-backed crash-atomic database. Recorded logical events replay deterministically; fresh model outputs and timing measurements need not.

4 SYSTEM AND INTERFACE CONDITIONS

The backend has asynchronous task-worker and replay adapters using the same commands. Three independently prompted workers share the existing GPU service. The server processes model requests serially under its configured capacity, while worker waiting, request delivery and user interaction proceed asynchronously. An agent here is a task worker, not a separate model architecture.⁴

The browser polls public state and updates pending work without replacing the active answer, source or text controls. It refuses older state sequences and

protects against delayed responses from a prior session. User actions and displayed request/version identities are logged. A deferred review restores its note and source. Saved sessions can be restored without launching workers. The ordinary interface never loads reference annotations.

Condition A presents worker conversations with unread indicators, navigable threads and source access. B presents a central conventional queue with one stable active review. C adds optional bounded source sessions, prominent pending-work views and admission controls. All conditions share source access, exact-version approval, explicit release and deferral semantics. We do not make a baseline unsafe to demonstrate a benefit from the treatment.

The prospective primary contrast is B versus C. A is a contextual baseline. C is a workflow bundle rather than an isolated batching mechanism. The software comparison measures its behavior under declared scripts; it does not estimate how people choose groups, defer work or use the pause control.

5 REAL-DATA EVALUATION DESIGN

5.1 Sources and Generation

TAT-QA provides real financial-report tables and text with human-written questions and reference annotations (Zhu et al., 2021). Worker assignments, source bundles, arrival schedules and reviewer scripts are constructed for this study. This is not a field study

⁴AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

of financial analysts.⁵

We froze the first 24 eligible unused public test-gold contexts in source order. The audit excluded all 36 contexts used in prior development or evaluation, matching normalized tables and long paragraphs, and detected cross-split source duplicates. Selected contexts have four to eight questions and a source rendering within the existing context limit. All 145 original questions are retained. There are no reliable complete report identifiers, so source contexts cannot be assumed to represent independent reports or companies. Public benchmark exposure to the model is not ruled out.

Two replicas produce 290 initial answers. The short interface requests an answer, scale and optional explanation or citations; it does not require an executable computation schema. The model may select a bounded arithmetic expression for a calculator to evaluate. No reference operands or answers are inserted. Four predetermined source rechecks produce actual revision outputs. Prompts, source selection, generation settings, failure handling and scenarios were committed before fresh generation.

The service is pinned Qwen2.5-7B-Instruct in BF16 on one RTX 6000 Ada, with zero CPU offloading. Temperature is 0.3, top- p is 1, and output allowance is 230 tokens within a 2,048-token model context. Seeds are recorded. The six development calls, 290 initial answers and four revisions consume 300 attempts. No request is replaced because it fails or produces an inconvenient answer.

5.2 Matched Software Scenarios

Twelve constructed bundles each contain two source contexts. Both replicas run in A, B and C at lower and higher arrival concentration, giving 144 scenarios. Every task is offered at time zero. Starts become eligible at intervals of 0.15 or 0.015 seconds, followed by a 0.04-second delivery delay. At most three replay jobs are in flight. These are compressed wall-clock test schedules, separate from observed GPU latency. They are not assumptions about human reading speed.

A script selects one request, waits 0.095 seconds and performs a question-specific ideal inspection. It approves a correct answer or creates an annotation-based correction, then explicitly releases that version. The first request is deferred once for 0.35 seconds. C pauses new starts from 0.25 to 0.65 seconds and may retain up to three source-related questions. Actual saved rechecks enter the declared replica-zero scenarios at 0.8 seconds. Every scenario ends at 2.2 seconds.

⁵AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

Methods share the saved answers, sources and eligible-start schedules. C's pause can change actual arrival times, but never the offered denominator. Every action and failure is retained. The unit for paired summaries is a source bundle, after averaging generation replicas. A fixed-seed bootstrap resamples 12 matched bundles 2,000 times. Such intervals describe the constructed scripts and do not estimate human effects. A separate live demonstration holds one review while real answers arrive and tests a version change without disclosing annotations.

6 RESULTS

6.1 Answer Quality and Scripted Completion

The initial model achieves 61/290 exact matches (21.0%), a mean token F1 of 22.61%, and 59/290 joint answer/scale matches. There are 38 empty answers. All returned bodies are parseable JSON. Optional citations supplied in a string instead of a list trigger 157 legacy formatting flags without removing the readable answer or source. This distinction matters when interpreting representation validity.⁶

All conditions release all 290 answers per load after ideal scripted correction (Table 1). The script's capacity is ample at the final cutoff. B and C therefore tie on completion in every paired bundle, with a degenerate script interval of zero. This is a mechanical result, not human equivalence. The low initial model accuracy makes the distinction between generated work and ideal correction particularly important.

C does not improve throughput. Under lower arrival concentration, its pause is followed by a larger queue and longer arrival-to-selection times. At higher concentration, all configurations reach similar queue peaks. These results expose the cost of pausing rather than assuming that a smaller visible queue is a benefit. Figure 2 includes work that has not started during the pause.

6.2 Concurrency, Versions and Latency

All 13,236 scenario events replay to their recorded logical states. The 144 cutoff snapshots and released-answer scores are reconstructed independently. There are no missing, duplicated or misassociated received

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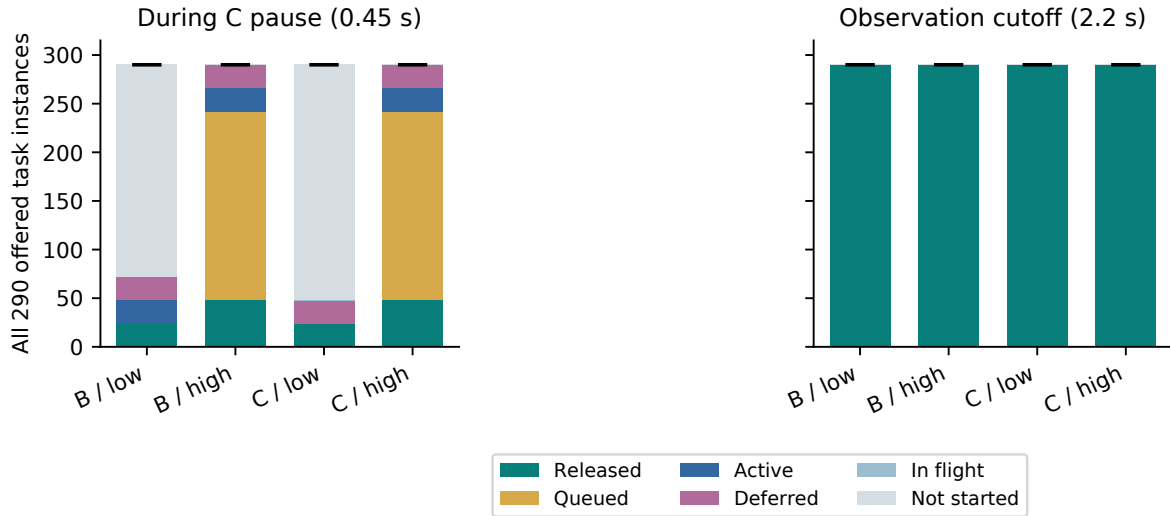


Figure 2: Recorded task accounting during the declared pause and at the final cutoff. The intermediate checkpoint is descriptive. Real model answers and source questions are replayed with constructed timing and ideal scripted corrections. Pausing shifts tasks into the unstated category without removing them.

Table 1: Measured scripted outcomes on 24 source contexts and two replicas. Peak is the mean sampled queue maximum. Wait is arrival-to-selection time, including resumed reviews. All releases use ideal simulated supervision.

View	Load	Released	Peak	Wait (ms)
A	lower	290/290	1.04	57.2
A	higher	290/290	10.08	573.6
B	lower	290/290	1.00	57.4
B	higher	290/290	10.08	572.8
C	lower	290/290	2.08	116.4
C	higher	290/290	10.08	583.5

request IDs in the completed matrix and no unexpected command failures. All 41,722 repeated active-snapshot checks preserve the reviewed output. These are software checks, not independent sources or participant observations.

The live demonstration uses twelve of the initial answers from two source contexts. A six-second scripted active review overlaps observed arrivals while new task starts are paused for two seconds. In-flight generation continues. A fresh model revision creates a new version, and the earlier approval attempt is refused. The demonstration retains all twelve unresolved requests rather than claiming a person completed them (Figure 3).

Across local measurements, median atomic handler time is 0.74 ms and its 95th percentile is 1.25 ms. Median browser HTTP round trip is 5.75 ms, with a 95th percentile of 15.64 ms. State receipt to JavaScript DOM update takes a median 1.60 ms and a 95th percentile of 6.12 ms. The latter excludes compositor paint. Delivery lateness relative to replay

timers has a 95th percentile of 2.17 ms. These boundaries exclude human response and GPU generation time.

Chromium tests pass 62 checks across all interfaces, including typed notes and focus during arrivals, source grouping, deferral, resumption, state-state rejection, explicit release, connection recovery and a timed study cutoff. Figure 4 shows the source-session component. The failure record includes early harness errors, a private annotation-lookup defect, a journal-restoration defect and a source-session UI timing defect that were repaired without replacing sources or model outputs. The implementation also received concurrency and snapshot-accounting fixes. These repairs and the original declaration are preserved.

7 PROSPECTIVE HUMAN EVALUATION

The next study compares B and C under two arrival concentrations within participants. Four disjoint six-context packets are balanced using public source length and question count. Williams condition orders cross packet rotations to avoid repeated answers within a participant. All conditions retain identical evidence, approval semantics and per-question authority. Training uses previously inspected development material outside the scored packets.⁷

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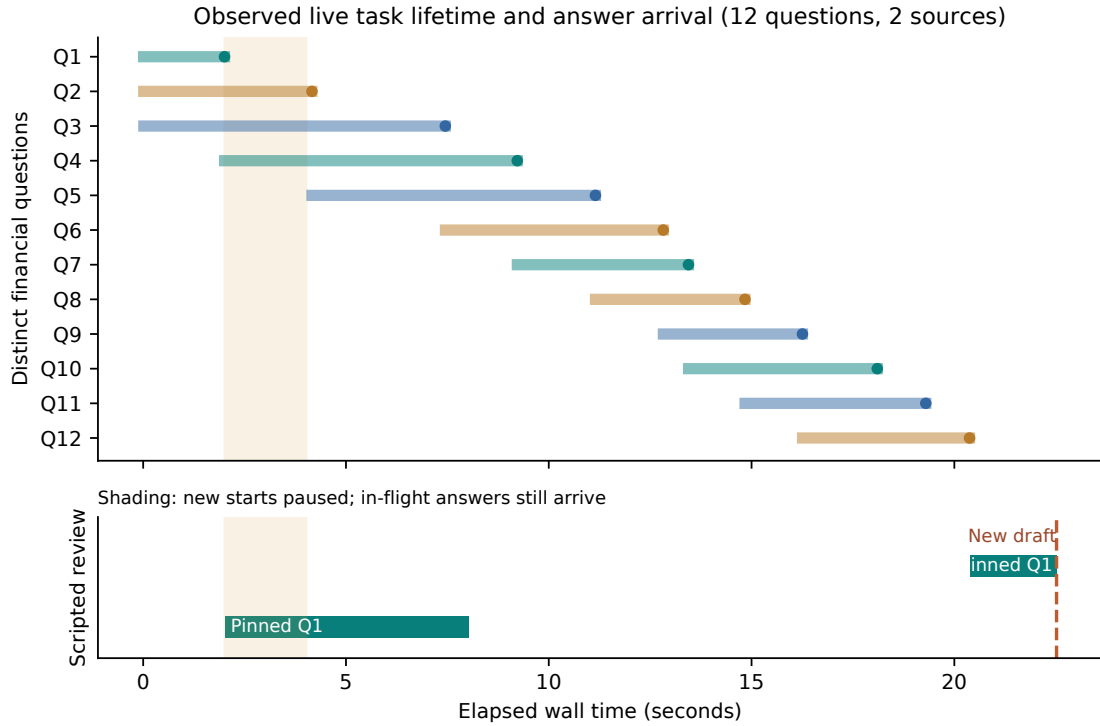


Figure 3: Observed task lifetimes and answer arrivals from actual GPU workers. Lifetimes include waiting for the shared serial model service. The lower panel records scripted review and revision actions. No duration in this figure is measured human review time.

The primary outcome is the proportion of all offered questions ending with a correct current-version release in a fixed session. Incorrect approvals, incorrect releases and unresolved work are reported separately. Rejection is not successful completion. Interaction measures include deferral, source revisits, active review intervals and resumption-to-decision time. Raw NASA-TLX items measure reported workload (Hart and Staveland, 1988); custom perceived-control items are retained separately rather than presented as a validated combined scale.

Analysis must account for both participants and source contexts. Clicks, questions and replicas cannot be treated as independent people. A formative pilot should check timing, accessibility, packet difficulty and the variance of paired outcomes. No powered sample size is claimed without those observations and a justified meaningful difference. Ethics review, consent and recruitment authorization remain necessary before involving participants. The supplied materials do not substitute for those steps, and no model-generated users are counted as participants.

8 DISCUSSION AND CONCLUSION

The workflow makes concurrent outputs addressable objects with explicit review and release authority. Its contribution is operational: a user can keep an answer in focus, inspect its original evidence, postpone it without losing context and see what remains before choosing the next task. The implementation and real-data traces make these behaviors inspectable and reproducible.⁸

The evaluation also limits the argument. Standard queues and version checks are strong predecessors, and the scripted experiment does not establish a session advantage. Ideal corrections turn weak model answers into perfect released outputs under an artificial service time. Source grouping may reduce repeated inspection, but it may also distract users or delay unrelated work. Strict version checks may require another review even when a regenerated answer appears unchanged. These tradeoffs require interaction observations.

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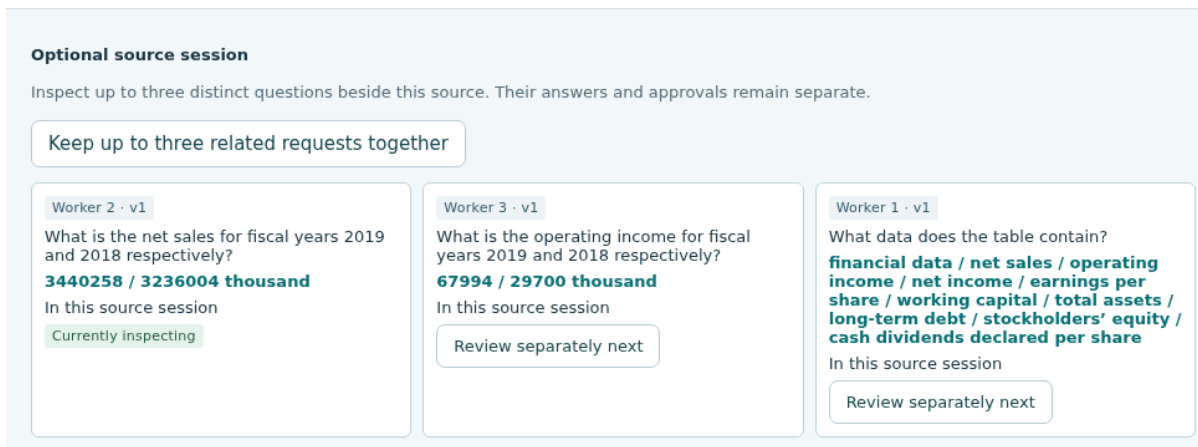


Figure 4: Actual interface capture during scripted replay. Three original questions share one financial source, while their generated answers and review decisions remain separate. The first selected source and first three questions determine this example. The long descriptive answer is retained despite imperfect benchmark agreement. This is software interaction, not a participant observation.

We therefore recommend the system as a position for testing user-controlled review, not as an established workload-reduction technique. The smallest next step is the prepared formative B/C study, followed by a prospectively specified effectiveness comparison. More elaborate automatic allocation is not required to investigate whether people can review more correct work while retaining control.

AI ASSISTANCE AND REPRODUCIBILITY

OpenAI Codex assisted with implementation, analysis, testing, figures and manuscript drafting (OpenAI, 2026). It is not an author or a participant. Human authors remain responsible for verification and submission. The separate research package contains frozen sources and settings, raw generations, private evaluator projections, event traces, prospective study materials and one-command saved-output reproduction. Historical studies and their manuscripts remain unchanged. No participant study, public submission or push was performed in this stage.

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